

HBS LAB Undergraduate Research Final

A Study in Mobile Robotic Hands and TCP Pulse Testing

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Abstract

This paper focuses on the development, fabrication, and testing of mobile robotic hand orthoses using TCP muscles. Over the course of the second half of the semester, my team and I conducted pulse testing experiments to train and test twisted and coiled polymer (TCP) muscles to be used in prosthetic humanoid robot hands. The first weeks were dedicated towards learning how to use the Tracker software and set up the training system. During the muscle training process, we collected data regarding muscle displacement at different inputs, such as actuation current, actuation time, and cooling time. This allowed us to better understand the TCP muscle characteristics. We also developed tests to measure the horizontal displacement needed to actuate our prototype finger and tested the muscle in line with a spring.

After the training, we began the main pulse testing of the TCP muscles under different electrical conditions. We ran currents from 2.5 A to 14 A at different time durations and measured the

maximum voltages for each trial. We optimized our setup by placing the spring, finger, and muscle all in line with one another and identified the best actuation and cooling times: 12 A for .25 seconds with a 10 second cooling time. We emphasized lowering the cooling time as much as possible to reduce energy and power usage, and found that it was difficult to reduce the cooling time below 10 seconds. These findings contribute to the development of more efficient, reliable prosthetic fingers actuated by TCP muscles.

Keywords- Twisted Coiled Polymer, Pulse Testing, Robotic Hand Orthosis

Midterm Summary

The focus of this research involves developing a low-cost, efficient humanoid robotic hand actuated by TCP muscles. Prior to the midterm, I studied existing literature such as research papers regarding TCP-actuated humanoid hands, SMA-actuated soft robotics, and various examples of existing hand orthoses. In addition to this

literature review, I also participated in hand-on lab work, such as TCP muscle fabrication, pulse testing, and displacement tracking. I also researched mobile applications, like Blynk and Arduino, for wireless control of the hand. Through these literature reviews and lab work, I began to explore the functionality and fabrication of TCP muscles and considered implementation into our robotic hand.

Week 11

Week 11 was dedicated towards initiating training of the TCP muscles and analyzing their displacement using video tracking software. We also devised a testing plan to evaluate actuation in the following week.



Figure 1. Training process using the Power Supply TCP Training set up

Abhishek taught me how to download the tracker software from Open Source Physics (OSP) as well as the corresponding Java download. He

provided me with a fabricated TCP muscle that had not been trained yet. I began the training process using the Power Supply TCP Training set up, as shown in Figure 1 above. I hooked the TCP muscle to a hook on the shelf and hooked a weight to the bottom of the muscle. I clamped one red wire to the top of the muscle and one to the bottom. We then hot glued a ruler besides the muscle and set up my phone to record a couple cycles of the TCP training process. To train the muscle, we used 2.3 A for 10 seconds on and 35 seconds off at 4V. We ran 5 cycles. We then ran the video through the tracker software.

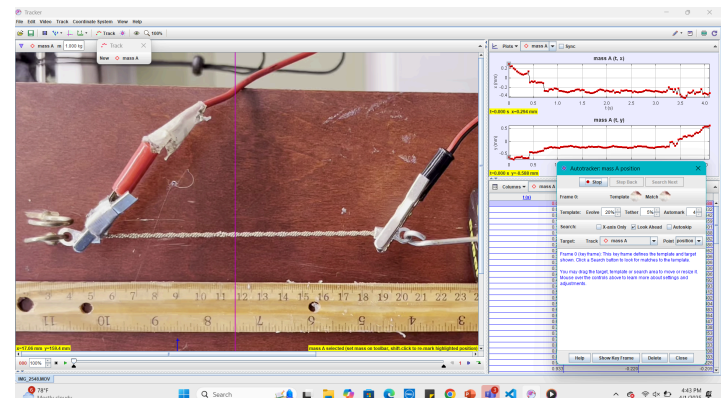


Figure 2. Using Open Source Physics Tracker to graph muscle displacement

Here is the procedure to use the tracker:

1. Import video.
2. Click on video, new filter, rotate if necessary.
3. Calibrate the calibration stick based on 1 cm on the ruler (set as 10 mm in software).

4. Set coordinate system to set up the origin.
5. Track mass A at the base of the origin and set mass there.

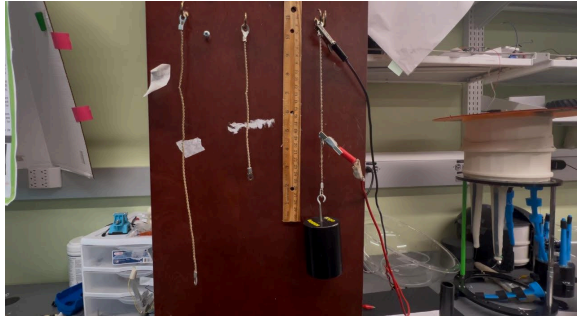


Figure 3. Set up for TCP muscle training

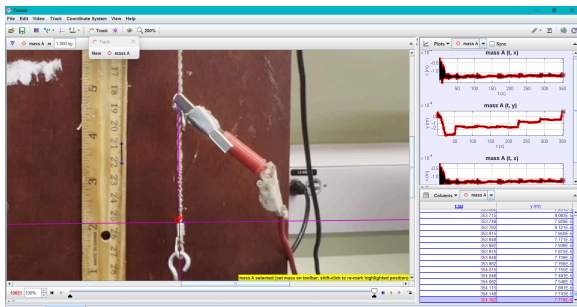


Figure 4. OSP Physics Tracker results- 45 cycles at 2.2A and 3.6 V

As seen in figure 4 above, I trained another muscle for 45 cycles at 2.2A and 3.6 V using the set up in figure 3.

Most of our time was spent waiting for the muscles to train. In the meantime, Joann and I devised a testing plan for next week with the help of Abhishek, including three tests.

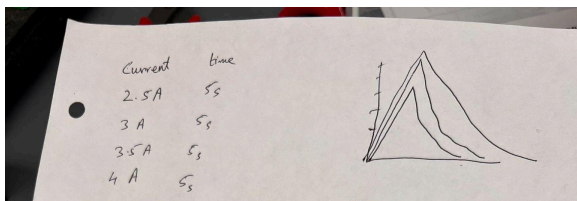


Figure 5. Muscle Displacement Trials

First, we would use the Power Supply TCP Training set up to test the muscle displacement. We would test at 4 conditions. 2.5A, 3A, 3.5A, and 4A, all at 5 seconds, as seen in Figure 5 above.

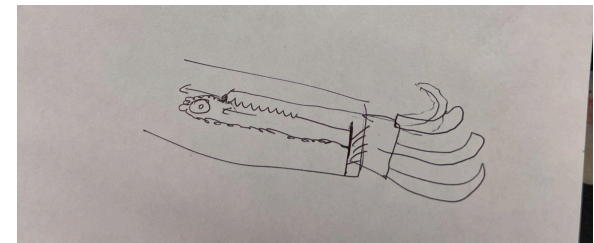
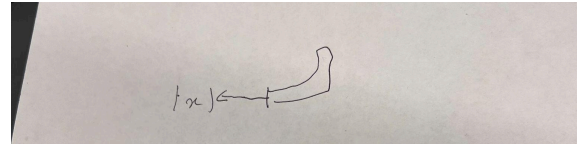


Figure 6. Linear Displacement TCP testing

Second, we would measure the linear displacement needed for the TCP muscle to fully bend the fingers in the prototype hand.

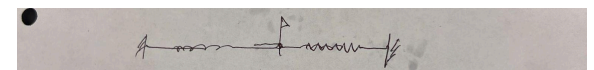


Figure 7. Linear Displacement TCP testing with spring

Lastly, we would measure the stretching of the spring in the following set up: we hook the TCP muscle to a fixture on the left, and then hook the other end of the muscle to a spring or series of springs. The spring would then be hooked to another fixture on the right. A rough sketch of the setup can be seen in the figure above.

Week 12

Week 12 was dedicated towards conducting the tests planned during Week 11.

Muscle Test 1:

Testing the muscle:

1. 2.5 A, 5 seconds, 0 A, 45 seconds
- 5 cycles
Max voltage: 8.5 V
2. 2.5 A, 1 seconds, 0 A, 45 seconds
- 5 cycles
Max voltage: 7.2 V
3. 2.5 A, 2 seconds, 0 A, 45 seconds
- 5 cycles
Max voltage: 7.5 V
4. 3.0 A, 5 seconds, 0 A, 45 seconds
- 5 cycles
Max voltage: 10.9 V
5. 3.0 A, 1 seconds, 0 A, 45 seconds
- 5 cycles
Max voltage: 8.7 V
6. 3.0 A, 2 seconds, 0 A, 45 seconds
- 5 cycles
Max voltage: 9.2 V
7. 3.5 A, 5 seconds, 0 A, 60 seconds
- 5 cycles
Max voltage: 13.6 V
8. 3.5 A, 1 seconds, 0 A, 60 seconds
- 5 cycles
Max voltage: 10.6 V
9. 3.5 A, 2 seconds, 0 A, 60 seconds
- 5 cycles
Max voltage: 11.6 V

10. 4.0 A, 2 seconds, 0 A, 90 seconds
- 5 cycles

Max voltage: 14.2 V

11. 4.0 A, 5 seconds, 0 A, 90 seconds
- 5 cycles

Max voltage: BROKE

The resulting graphs were constructed during Week 13.

Spring Test 2:

We fixed the spring in parallel with the finger to the TCP muscle. We fixed the spring to the wood plank and the finger base to the wood plank.

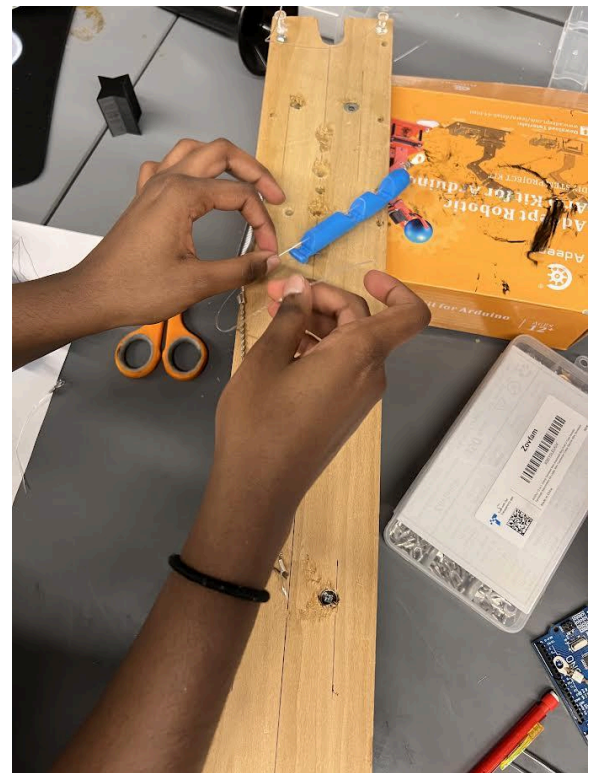


Figure 8. Joann and I constructing Test 2 set up.

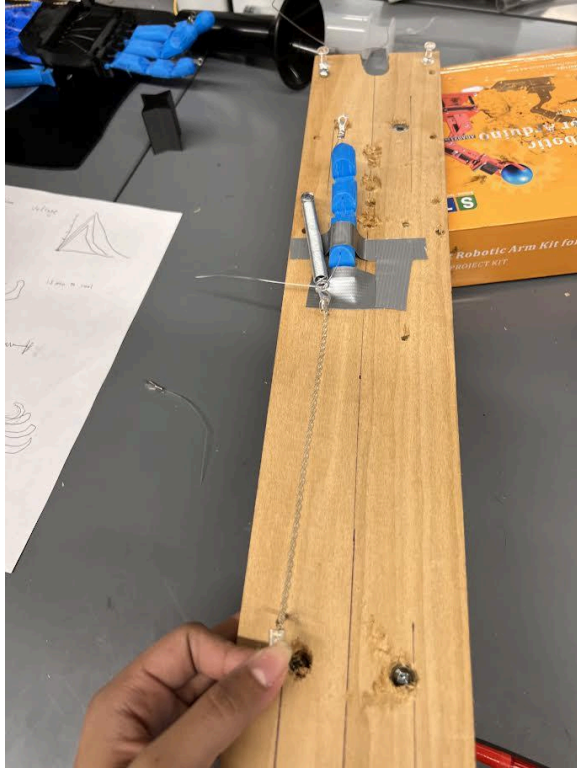


Figure 9. First attempt at Test 2 set up.

The first attempt at our set up can be seen in the image above. However, the spring needed to be placed vertically above the finger to allow for the spring to be pre-stretched when pulling the TCP muscle.

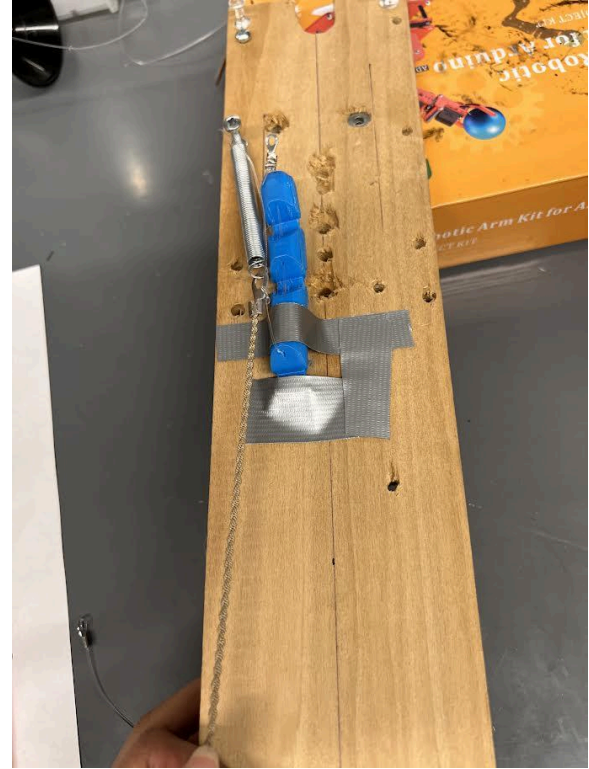


Figure 10. Second attempt at Test 2 set up.

The second, correct set up is shown in the figure above. Using this set up, we concluded that the horizontal displacement needed to completely bend the finger was 7.75 cm.

Week 13

During Week 13, I ran the videos from Week 12 experimentation through the OSP physics tracker software. We also conducted more pulse testing and graphed those as well.

Graphs from Week 12 Testing:

3.0 A, 2 seconds, 0 A, 45 seconds - 5 cycles

Max voltage: 9.2 V

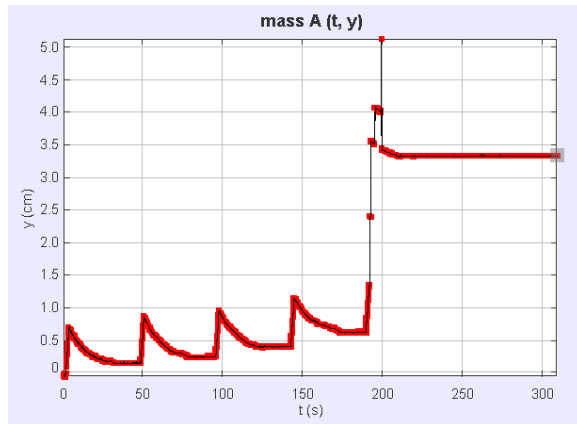


Figure 11. 3.0 A, 2 seconds, 0 A, 45 seconds - 5 cycles.

4.0 A, 5 seconds, 0 A, 90 seconds - 5 cycles

Max voltage: BROKE on first cycle

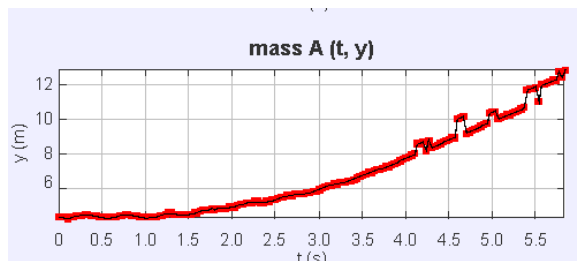


Figure 12. 4.0 A, 5 seconds, 0 A, 90 seconds - 5 cycles

4.0 A, 2 seconds, 0 A, 90 seconds - 5 cycles

Max voltage: 14.2 V

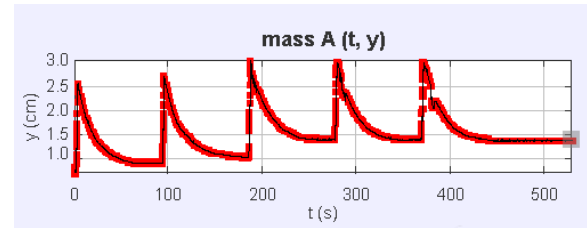


Figure 13. 4.0 A, 2 seconds, 0 A, 90 seconds - 5 cycles

3.5 A, 1 second, 0 A, 60 seconds - 5 cycles

Max voltage: 10.6 V

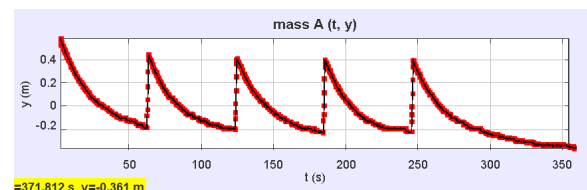


Figure 14. 3.5 A, 1 second, 0 A, 60 seconds - 5 cycles

3.5 A, 2 seconds, 0 A, 60 seconds - 5 cycles

Max voltage: 11.6 V

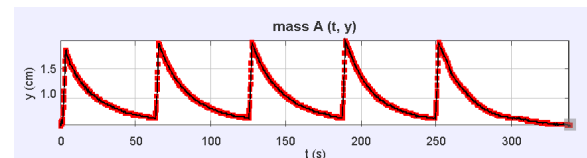


Figure 15. 3.5 A, 2 second, 0 A, 60 seconds - 5 cycles

We then began new pulse testing. We actuated a new TCP muscle at 2.2 A for 5 seconds, for 10 minutes to train the muscle.

Further Testing:

.25 S, 5 A, 0 A, 120 seconds - 5 cycles
Max voltage - 13.1 V

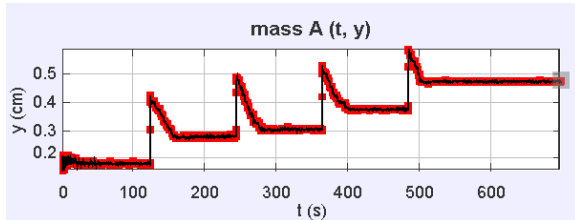


Figure 16. 5 A, .25 seconds, 0 A, 120 seconds - 5 cycles

.25 S, 6 A, 0 A, 120 seconds - 5 cycles
Max voltage - 16.1 V

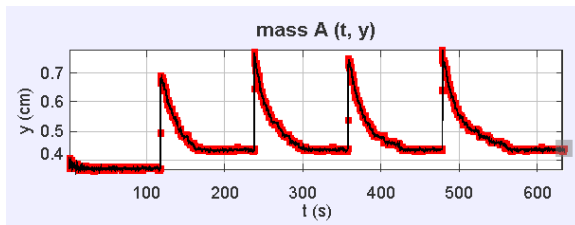


Figure 17. 6 A, .25 seconds, 0 A, 120 seconds - 5 cycles

.25 S, 7 A, 0 A, 120 seconds - 5 cycles
Max voltage - 17 V

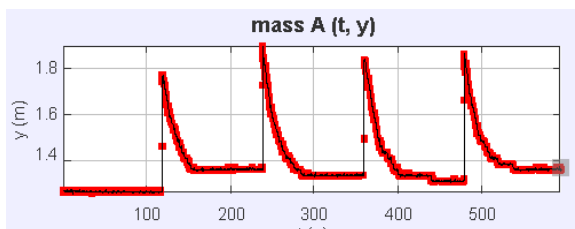


Figure 18. 7 A, .25 seconds, 0 A, 120 seconds - 5 cycles

This allowed us to determine that we could safely keep increasing the amplitude for increased angle of finger deflection. We continued to push this limit in Week 14.

Week 14

During Week 14, we were tasked with testing the maximum current we could actuate the muscle with for various durations before the muscle would break.



Figure 19. Original set up to see finger displacement when actuated

Using this set up, we actuated the muscle to see if the current was sufficient. The length of the actuator was 165 mm.

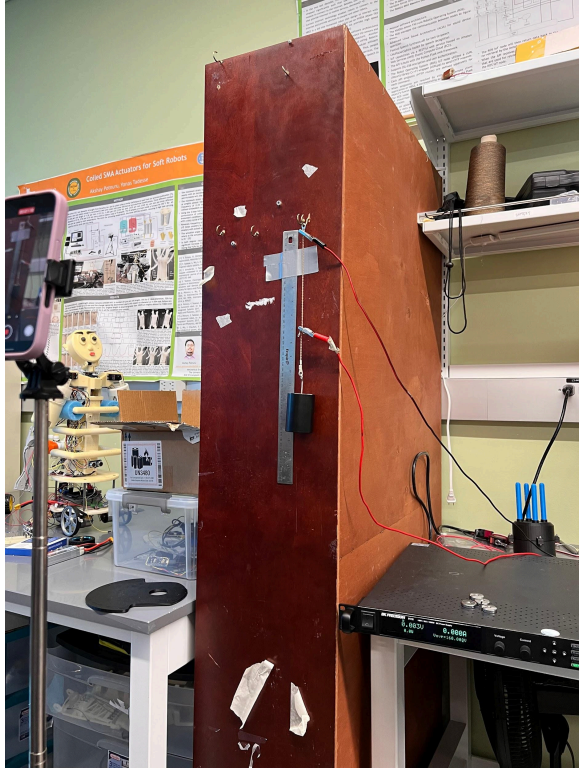


Figure 20. Pulse Testing Setup

Using the set up above, we tried to test when the muscle breaks by increasing the amplitude by .5 A each trial while keeping timing the same.

1. 9.0 A, .25 seconds, 0 A, 45 seconds - 5 cycles
Max voltage: 29.9 V
2. 9.0 A, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 17.3 V
3. 9.5 A, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 17.6 V
4. 10.0 A, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 19.7 V
5. 10.5 A, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 20.6 V
6. 11.0 A, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 22.3 V
7. 11.5, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 22.3 V
8. 12.0, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 22.3 V
9. 12.5, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 24.6 V
10. 13, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 28.0 V
11. 13.5, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 28.0 V
12. 14, .25 seconds, 0 A, 45 seconds - 4 cycles
Max voltage: 28.5 V

13. 14.5, .25 seconds, 0 A, 45 seconds
- 4 cycles
Max voltage: 29.2 V

However, the muscle did not break at any of these currents.

I also helped Pranita assemble the fingers into the hand by stringing the line through all of the fingers.

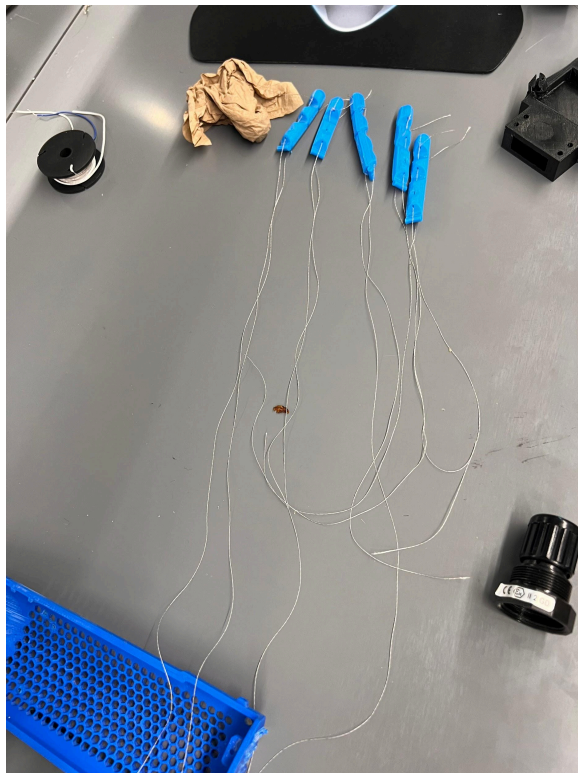


Figure 21. Fingers strung with fishing line

While Pranita and I were assembling the fingers, I also observed Joann as she built the circuit for one finger.

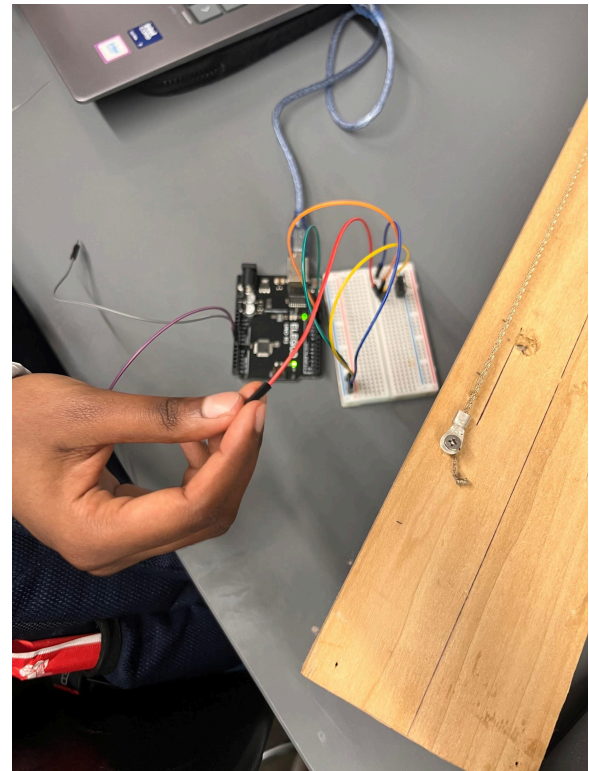


Figure 22. Circuit to actuate one finger

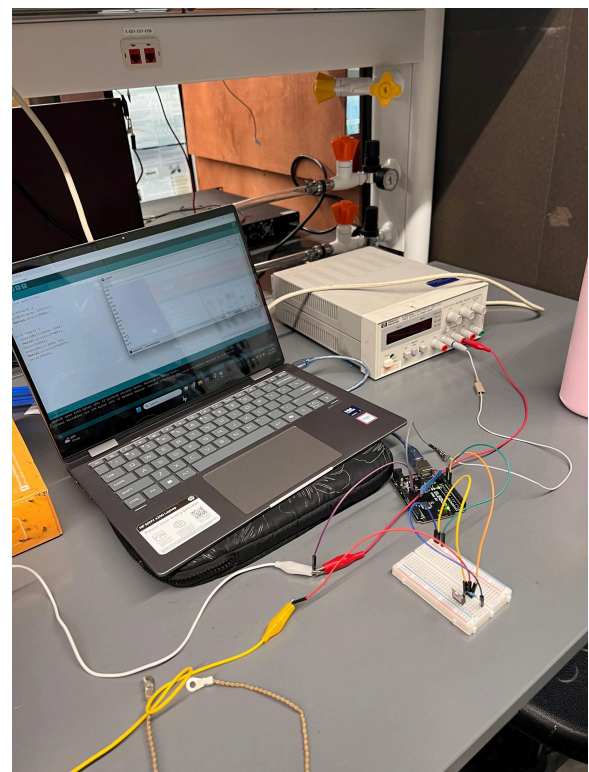


Figure 23. Arduino code for actuating the muscle

Week 15

During Week 15, we tested the deflection of the finger under various currents. We simultaneously tried to reduce cooling time and find the optimal combination of current and cooling time without breaking the muscle and while giving the finger sufficient time to return to its resting position after cooling.

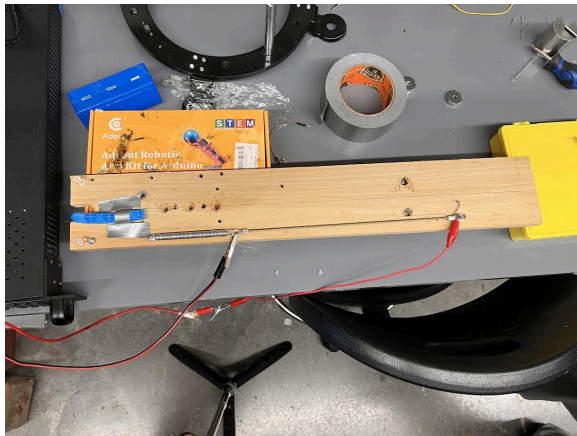


Figure 24. Nonoptimal finger testing set up

Originally, our finger was not placed in parallel with the TCP muscle and spring. What this meant was that the force of actuation was being applied to the finger at an angle, thus wasting some of the force's magnitude. To rectify this, we moved our finger over to be placed above the spring and tightened the muscle to pre-stretch the spring and allow for better actuation.

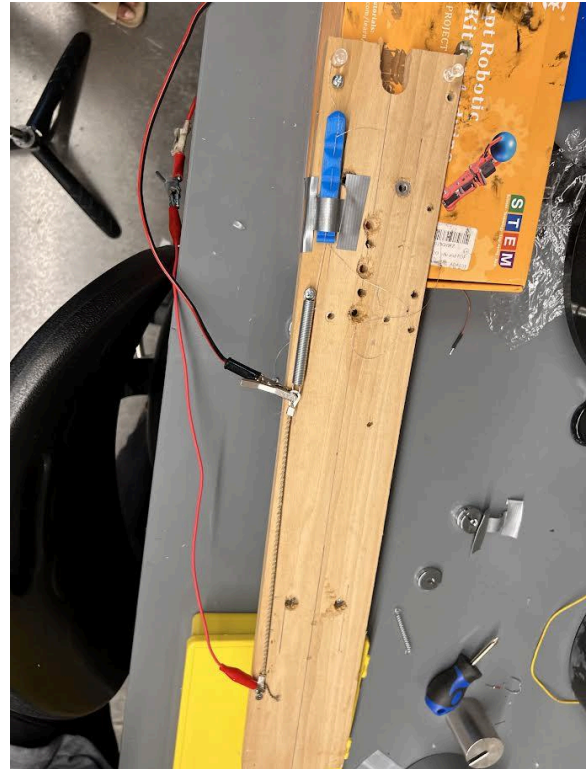


Figure 25. Optimized finger testing set up

This fixed set up can be seen in the photo above. Using this arrangement, we tested the best combination for high actuation current and lowest cooling time. We started at 10 A at .25 seconds and a cooling of 0 A at 90 seconds. We raised the amperes by an increment of .5 A for each trial until we got to 14A and ran our cooling consistently at 35 seconds. We eventually raised our actuation to 14 A and lowered our cooling time to 25 seconds. At this combination, the finger returned to its original position without trouble and bent a significant amount, showing us that 14 A was a sufficient current for actuation.

We identified the best combination for actuation current and cooling time was 14 A at .25 s and 0 A at 25 s.

Week 16

During Week 16, we were instructed to determine the lowest possible cooling time with sufficient current for actuation. This would allow for a faster frequency of finger movement. We conducted the following trials:

8A, 1 second, 0A, 5 seconds -> muscle broke

8A, .125 seconds, 0A, 5 seconds -> not sufficient actuation

12A, .125 seconds, 0A, 5 seconds -> not sufficient actuation, as actuation was building off past actuations

8A, .25 seconds, 0A, 5 seconds -> not sufficient actuation, as actuation was building off past actuations, not sufficient cooling

8A, .25 seconds, 0A, 10 seconds -> not sufficient actuation, as actuation was building off past actuations

9A, .25 seconds, 0A, 10 seconds -> not sufficient actuation, as actuation was building off past actuations

10A, .25 seconds, 0A, 10 seconds -> not sufficient actuation, as actuation was building off past actuations

10A, .125 seconds, 0A, 10 seconds -> actuation was better but could still be improved

12A, .125 seconds, 0A, 30 seconds -> max 30V

sufficient actuation, we wanted to see if we could reduce cooling time further

12A, .125 seconds, 0A, 10 seconds -> best combination, max 30V

Graph:

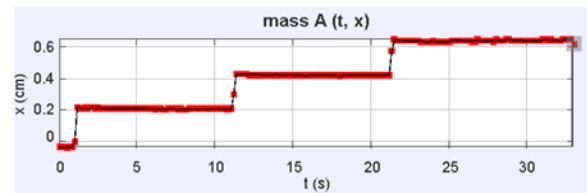


Figure 26. Optimized actuation and cooling combination

16A, .125 seconds, 0A, 5 seconds -> smoke, ended trial early

We concluded that 12A at .125 seconds coupled with cooling of 0A at 10 seconds seems to give the best tradeoff between finger actuation and cooling time. Increasing the current past 14 A is not advised as per our previous tests, as it often results in smoke or broken muscles.

Conclusion:

During the second half of this semester, my team and I made significant progress in investigating the performance and characteristics of TCP muscles for actuating humanoid robotic fingers. We began our research by training the TCP muscles and varying our input currents, actuation times, and cooling times to observe the muscle behavior. And through a series of pulse tests over the course of a few weeks, we first identified 14 A at 0.25 seconds actuation with 25 seconds of cooling as the optimal combination for sufficient angle of finger deflection as well as sufficient cooling time for the finger to return to its original position. However, we were asked to further reduce the cooling time to allow

for a faster frequency of finger movement. Further testing in Week 16 showed that higher currents (above 14 A) caused excessive heating or smoke, while lower durations of cooling time led to incomplete muscle recovery back to original positioning. Ultimately, we concluded that 12 A at 0.125 seconds with a 10-second cooling interval offered the best balance between performance and cooling time, providing effective and frequent actuation. Our findings support the design of more efficient and responsive robotic hands using TCP muscles and can help future RIDE students further build off of our findings.